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Precision measurement of the neutron magnetic form factor

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Abstract

We present a precise measurement of the neutron magnetic form factor G_{mn} at low momentum transfer ($q = 1.69 \text{ fm}^{-1}$). From a simultaneous measurement of $D(e, e'n)$ and $D(e, e'p)$ we obtain the ratio of neutron and proton cross sections. The neutron detection efficiency is obtained from a separate measurement using tagged neutrons produced by $H(n, p)n$ scattering of a monochromatic neutron beam. In contrast to previous determinations of G_{mn} , the present value is insensitive to the systematic uncertainties in the interpretation of the data in terms of G_{mn} and represents a determination of G_{mn} to $\pm 1.7\%$.

1. Introduction

The accurate knowledge of the nucleon form factors is fundamental for an understanding of the nucleon in terms of QCD. At the same time, these form factors are needed for quantitative calculations of nuclear form factors.

Much of the data on the neutron magnetic form factor comes from experiments done more than two decades ago [1,2]. The values for G_{mn}^2 exhibit a scatter of typically $\pm 20\%$ over much of the range of momentum transfer q explored (for a compilation see [3]).

Experimental uncertainties account for only a small part of the large fluctuations in G_{mn} ; theoretical uncertainties in the interpretation of the data in terms of G_{mn} are by far the dominant source. In the absence of

free neutron-targets, the neutron form factor has been extracted from quasielastic electron scattering data on deuterium. Three methods have been employed:

- (i) Inclusive quasielastic scattering, $D(e, e')$. This requires a longitudinal/transverse separation, and a subtraction of the (dominant) proton magnetic contribution. The main systematic error results from the uncertainty of the deuteron wave function and the contribution of meson exchange currents (MEC), the effects of which are greatly amplified by the two subsequent subtractions.
- (ii) Quasi-coincidence experiments $D(e, e'\bar{p})$. Relying on the *absence* of a recoiling proton as identification of $(e, e'n)$ requires an excellent understanding of the reaction mechanism, which is not at hand, and on a perfect knowledge of the deuteron wave function.
- (iii) Coincidence experiments $D(e, e'n)$. This re-

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quires a neutron detector with a well known absolute efficiency, which was usually determined using the $\gamma + p \rightarrow \pi + n$ reaction. The extracted value of G_{mn} depends on the deuteron wave function assumed, and on MEC corrections (depending on whether the top of, or the full quasielastic peak is used). Such a measurement has been attempted 20 years ago at large q [2], and only recently at lower q values [4].

To avoid a number of the above problems, we employ a novel method that combines the *simultaneous* measurement of $D(e, e'n)$ and $D(e, e'p)$ with a precise determination of the neutron efficiency using a high intensity neutron beam. We exploit the fact that the ratio of cross sections for $D(e, e'n)$ and $D(e, e'p)$ directly measures the ratio of cross sections on the *bound* neutron and proton, which, for the case of deuterium, is close to the cross section ratio on *free* nucleons. This method avoids large subtractions, the effects due to the neutron and proton wave functions in the deuteron cancel as they are the same if the sampled phase space is identical, and the effects of n - p final state interaction (FSI) and MEC are nearly the same. The remaining theoretical corrections are small and sufficiently well understood. The neutron detection efficiency can be measured with high accuracy using high-intensity neutron beams available at modern hadron-beam facilities; this allows the neutron-detector response to be studied in the detail required for a *precision* measurement.

For the measurement of the required quantities, we have performed two separate electron scattering experiments, which were bracketed by three neutron scattering experiments for the determination of the efficiency, thus ensuring a thorough control over reproducibility of the results.

2. $D(e, e'n)$ experiment

The experimental set-up used for the electron scattering experiment, performed at the MEA accelerator of NIKHEF (Amsterdam), is sketched in Fig. 1a; a more detailed description of this experiment is given in Ref. [5].

The ratio $R = d\sigma(e, e'n)/d\Omega/d\sigma(e, e'p)/d\Omega$ on D was measured in quasielastic kinematics at four momentum transfers $q=1.56$ and 1.80 fm^{-1} , referred to

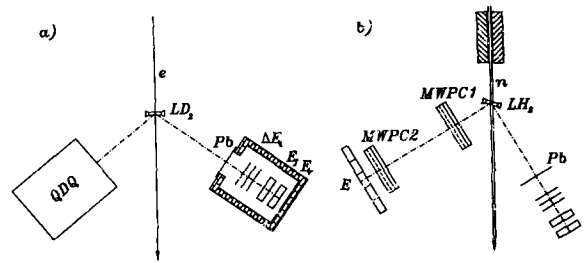


Fig. 1. Experimental set-up used at NIKHEF (a) and PSI (b).

as kinematics I and II. The average intensity of the electron beam was varied from 70 to 2000 nA with a duty cycle of 0.5 to 1%. The beam energy was 457.7 (450.9) MeV for kinematics I (II). The scattered electrons were detected in the QDQ magnetic spectrometer [6] within a momentum acceptance of 6% around the central momentum of 403.0 MeV/c (381.5 MeV/c) at scattering angles of 41.9° (50.8°). The target consisted of 3 mm of liquid deuterium contained in a cell with windows made of $2 \mu\text{m}$ Havar foils.

The recoiling nucleons were detected in a telescope consisting of three ΔE_i scintillators 1.2 (2.1) mm thick for kinematics I (II), followed by two scintillators, E_{front} and E_{rear} 50 mm thick (see Fig. 1). All scintillators were of dimensions $25 \times 25 \text{ cm}^2$. The distance from the target to the center of E_{front} was 174 cm. The geometry was such that the nucleon detector covered about 25% of the Fermi cone.

The nucleon detector was placed in a lead and concrete shielding house. A 1.1 (2.8) mm thick lead absorber for kinematics I (II) was placed at the entrance window of the lead house in order to absorb photons of very low energy. The thicknesses of the detectors were chosen such that protons from the $D(e, e'p)$ reaction could not reach E_{rear} . During the experiment, front and rear counters were regularly interchanged.

To achieve the desired experimental accuracy, a precise energy calibration of the recoil detector was necessary. Two temperature stabilized LED's were attached to each of the E-counters. Their light output was monitored by a PIN-diode, the long-term stability of which is known [7]. The LED's flashed at two different amplitudes at a rate of 1 Hz. The LED information was used mainly for relative corrections of the overall gain and pedestal, and resulted in a stability of better than 0.4% during the experiment. Protons from the reactions $D(e, e'p)$ and $H(e, e')p$ (on

the small admixture of hydrogen in the deuterium) were recorded continuously, and served as a source of events with known energy deposition. The uncertainty of the absolute energy calibration from these data was $\pm 0.15 \text{ MeV}_{ee}$ ($\pm 0.10 \text{ MeV}_{ee}$) for kinematics I (II).

Since the reactions $(e, e'n)$ and $(e, e'p)$ were measured *simultaneously*, the ratio R is independent of the integrated luminosity, dead time, inefficiencies in the detection of the electrons and, to first order, target thickness.

In the off-line analysis, protons were defined as events for which E_{front} and at least one ΔE counter had a correct time-of-flight relative to the electron. The detection efficiency for protons of counter ΔE_i was determined from the number of protons identified in ΔE_j and ΔE_k but not seen in ΔE_i . It varied between 85% and 99% depending on i and the software threshold used. The proton yield, corrected for the efficiency of the ΔE detector used, was determined independently for each of the three ΔE detectors ($i = 1, 2, 3$). The resulting three values agreed to within 0.1%, and the average value was taken for the final result.

Neutrons were defined as those events which had a correct time signal in E_{front} and in at least two ΔE counters no signal above a given software threshold value. With such a definition a signal to noise ratio of up to 25 was achieved. Inefficiencies of the ΔE counters lead to protons in E_{front} being misidentified as neutrons. The fraction of misidentified protons was determined by exploiting the time correlation between E_{front} and the remaining ΔE counter and led to a 2–4% correction of the neutron flux.

The pile-up probability was deduced by comparing the number of hits in the time-of-flight peak of E_{rear} , with and without the neutron identification requirement in the ΔE counters. After all corrections, the number of neutrons was independent of the defining ΔE pair to within 0.2%.

3. $H(n, p)n$ experiment

For the calibration of the neutron detector, we performed experiments with a *tagged* neutron beam, using the neutron beam facility of the PSI injector cyclotron in Villigen. The set-up used is shown in Fig. 1b, the calibration experiments are described in more detail in Ref. [8].

The neutron facility at PSI [9] provided an essentially monoenergetic, well collimated neutron beam of 50 and 68 MeV. The flux of neutrons amounted to $10^6/\text{s}$ for a beam diameter of 2.7 cm. The neutrons were elastically scattered from a target of liquid hydrogen of 0.7 cm thickness (small contributions from the target windows were measured and subtracted). The recoiling protons were detected using a thin plastic counter ΔE , two sets of multi-wire proportional chambers (MWPC) each measuring the x - and y -coordinates with 2 mm resolution, and a thick, segmented E plastic counter. With this set of detectors, the proton time-of-flight relative to the cyclotron RF, the proton x_i and y_i coordinates at the target, its direction in θ, ϕ , and its energy were measured. From this information, the trajectory and energy of the scattered neutron could be derived accurately, and the precise number of neutrons incident on the (central quarter of the) neutron detector could be specified without knowledge of the incident neutron flux, the target thickness, the n - p cross section, the proton detection efficiency and counter dead times.

The neutron detector, in the *identical* configuration as used in the $(e, e'n)$ experiment, was placed in the 'beam' of tagged neutrons. The number of detected neutrons was determined as the number of counts with a hit in E_{front} above a given software threshold value and at least two ΔE counters not being hit. Due to the much lower background as compared to the $(e, e'n)$ -experiment corrections due to misidentified protons or pile-up were negligible. Use of the identical configuration ensured that absorption effects of neutrons were automatically included in the efficiency determined.

While for kinematics I we have calibrated the detector with tagged neutrons of the exact (average) energy corresponding to the $(e, e'n)$ -run, the *absolute* calibration for kinematics II was performed in a different way: Here, the (relative) dependence of the calibration on neutron energy was measured by placing the detector in the neutron beam, and exploiting the precisely known relative energy-dependence of n - p scattering under specially selected kinematics.

During the calibration runs, the gains of the various photomultipliers were measured with the same LED/PIN system employed during the $(e, e'n)$ runs. Monoenergetic protons recoiling from the LH_2 target, collimated to strike a well defined area of the neutron detector only, provided an independent absolute

Table 1

Experimentally determined corrections together with uncertainties in the determination of R . I and II denotes the two measured kinematics.

Correction and error in % of R	I	II
Statistical error of the yield ratio	± 2.7	± 1.5
Total error in neutron efficiency	± 1.2	± 1.7
Illumination and resolution correction	9.2 ± 0.5	11.3 ± 0.5
Proton losses due to MS and absorption	10.9 ± 0.9	20.7 ± 1.1
(n, p) -conversion	< 0.005	< 0.005
(p, n) -conversion	0.22 ± 0.05	0.52 ± 0.13
Contribution of target windows	0.20 ± 0.02	0.20 ± 0.02
Gain stability $\pm 0.4\%$	± 0.28	± 0.36
Energy-loss difference PSI/NIKHEF $\pm 0.1 \text{ MeV}_{ee}$	± 0.70	± 0.90
Energy calibration at NIKHEF	± 1.1	± 0.9
Threshold dependence of R	± 1.3	± 0.0
Total uncertainty of R	± 3.6	± 2.9

energy-calibration. For the threshold values used in the analysis of the $(e, e'n)$ runs, 10 MeV_{ee} (16 MeV_{ee}) for kinematics I (II), we determined the efficiencies to $5.10 \pm 0.06\%$ ($3.75 \pm 0.06\%$). The error includes both statistical and systematic error added in quadrature.

The calibration with tagged neutrons provides an *absolute* measurement of the efficiency, albeit for a specific distribution of events in x , y over the detector, and a specific distribution in energy around the mean value of interest. For a precise calibration, it is necessary to correct for the fact that these distributions are not equal in the $H(n, p)n$ and $D(e, e'n)$ experiments. These corrections need to be quantitatively understood; edge effects (up to 10%) are important because the neutron detector intercepts only part of the Fermi-cone in the $(e, e'n)$ experiment (an unavoidable situation if the theoretical FSI and MEC corrections are to be kept small). The corrections due to the various effects, which were neglected in [4], were measured by placing the neutron detector directly in the neutron beam and scanning its response in x and y . The use of a collimated, monoenergetic neutron beam of high intensity allows for a detailed study of the relative efficiency as a function of x and y . The ratio of appropriately weighted relative efficiency distributions normalized to the absolute calibration measurements allowed for a precise determination of the average efficiency. These corrections in R due to the different illuminations and resolution functions are listed in Table 1.

Placing the MWPC in front of the neutron detector allowed the measurement of a number of corrections for protons as a function of x and y . Measurements with and without absorbers in front of the neutron detector provided information on the absorptive losses, (p, n) -conversion and corrections for multiple scattering (MS). These corrections could be determined with high accuracy as they are only dependent on ratios of rates which can be measured with good statistics given the high n , p -fluxes. The (p, n) -conversions are determined in an analogous way from the ratio of incident protons to the number of detected neutrons in the detector. The results of these studies are also listed in Table 1.

4. Evaluation of R

The results of the two experiments, the yield ratio of detected neutrons and protons and the response of the detector, allow the determination of R , the ratio of the e - n and e - p cross sections. The determined values of R , 0.110 ± 0.004 (0.154 ± 0.004) for kinematics I (II), are found to be independent of the cut-off energies employed which confirms consistency of the data. The uncertainties contributing to the error of R are listed in Table 1.

In order to proceed from the cross section ratio R to G_{mn} , we have to correct for the effects beyond PWIA. The nuclear effects have been calculated in two theoretical approaches. The calculation of Arenhövel [10]

includes π , ρ and ω exchange, and $\Delta\Delta$ Isobar Currents (IC), whereas Hummel and Tjon [11] consider only π and ρ exchange and no IC. The n - p final state interaction, mostly charge exchange scattering, gives the largest contribution, and its effect on R is calculated quite consistently in the two models: Arenhövel [12] finds FSI corrections of -9.0% (-6.1%), whereas Hummel and Tjon [13] obtain corrections of -11.2% (-8.0%), for kinematics I (II). For the evaluation of R_{PWIA} , we use the average of the values quoted with an error bar of $1/2$ their difference. The corrections due to MEC are estimated in both models, and for both kinematics, to be 1% , whereas the influence of IC, calculated only by Arenhövel, amounts to 0.6% . The effect of the different effective solid angles of proton and neutron on the corrections, determined by integrating the theoretical cross sections over the experimentally accepted phase space, is negligible ($< 0.1\%$).

The contribution of the neutron electric form factor to R is small (1%); the corresponding correction introduces only a small additional uncertainty despite the poor knowledge of G_{en} .

5. e - p cross section

The basis of the method used here is the measurement of the e - n cross section (which is essentially given by G_{mn}) relative to the e - p cross section which we derive from the world's supply of e - p cross sections [14]. To obtain a precise value of σ_{ep} at the q desired, we have considered all data in a range of 0.4 fm^{-1} around this value; no rapid q -dependence, apart from the general fall-off, is expected. In this range of q we have used a parameterization with a relative q -dependence as given by the standard global parameterizations of G_{ep} and G_{mp} , with the overall normalizations fitted to the data. The statistical errors of the data have been treated in the standard way, while the systematic errors have been treated by changing the data of each set by their error, and by refitting. The effects of the systematic errors of different experiments have been added in quadrature and yield a total error of 2.0% .

We find that the proton cross section σ_{ep} is 6.6% (7.9%) below the value given by the standard Dipole parameterization ($M_D^2=0.710 \text{ GeV}^2/c^2$) for kinematics I (II) of the ($e, e'n$) experiment. The fitted values

for σ_{ep} are 823 nb/sr (340 nb/sr).

6. Conclusion

With the experimental values of σ_{ep} and the measured values of R_{PWIA} we determine the ratio of G_{mn} to the 'standard' value given by the Dipole parameterization ($G_{mn}/\mu_N G_D$) to 0.945 ± 0.020 (0.967 ± 0.018) for kinematics I (II). The combined error includes all experimental errors the errors of the theoretical corrections and the error of σ_{ep} added in quadrature. As the values of ($G_{mn}/\mu_N G_D$) determined for the two neighbouring momentum transfers are consistent, and ($G_{mn}/\mu_N G_D$) is not expected to show any rapid change we take a weighted mean. The uncertainty is calculated by taking into account that some of the systematic errors of the two determinations are independent. The combined result of ($G_{mn}/\mu_N G_D$) gives a value of 0.957 ± 0.016 for the median momentum transfer $q=1.69 \text{ fm}^{-1}$ of the present experiment.

There is a clear discrepancy between the present result and the one of Markowitz et al. [4]. The measurement of Ref. [4] has been analyzed under the assumption, that the neutrons hit only the central part of the detector and edge effects can be neglected for both the $D(e, e'n)$ and the $H(\gamma, n)p$ calibration experiment. While this is correct for the latter, the deuteron momentum distribution leads to an important spread in angle of the recoiling neutron in the former. As a consequence, the neutron intensity at the detector edges is down by only $\sim 35\%$ as compared to the center. The resulting edge effects which we experimentally find to be of order 10% in our detector of similar size (see third entry of Table 1) and contributions from collimator scattering are thus important and would need to be taken into account.

The precision of the result achieved in the present experiment, a 1.7% error in G_{mn} , shows that it is possible to obtain very precise neutron form factors when using an approach that depends the least on the theoretical input, and when performing the needed calibrations using modern high-intensity neutron beams.

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